ALIEN.AI

AI Image & Video Creation Platform

PRODUCT PRESENTATION • UI/UX SPECIFICATION • BACKEND & API PLAN

A step-by-step blueprint based on your uploaded Alien.ai website, screenshots, screen recording and existing project ZIP.

VISUAL DIRECTION	IMAGE ENGINE	VIDEO ENGINE	STARTER BALANCE
Deep black • purple glow • violet gradients • mint green highlights	OpenAI Images API	Runway API	50 credits per new account

Prepared from the supplied reference material • 05 November 2026

01 / PRODUCT VISION

Alien.ai is a single workspace where a user can create AI images, generate short AI videos, upload reference media, and return later to continue from their own saved history. The product should feel like a polished creative studio—not a static landing page or a demo-only UI.

Primary product goals

- Make the homepage visually consistent with the uploaded Alien.ai reference: dark base, violet/purple glow, mint accents, large hero headline, rounded CTA and featured media preview.
- After Get Started or Create, route the visitor to Signup / Login. A successful login opens that account’s History page.
- History is the user’s personal library. Create New opens the generation dashboard; previous assets remain available after refresh and on another device.
- Provide one prompt composer for both image and video generation, with upload, mode, aspect ratio, duration and Generate controls.
- Connect real provider APIs. Show queued / processing / completed / failed states and never show a fake success result.

Important product boundary

Use Google Flow as a workflow reference for the creator workspace (large canvas, prompt composer, media attachments, model/settings controls and generation history). Build an original Alien.ai visual system and interaction design; do not copy Google branding, proprietary assets or claim to be Google Flow.

02 / VISUAL SYSTEM & BRAND

The uploaded homepage screenshot is the visual source of truth for the public landing page. Keep the same overall palette and hierarchy while using an original Alien.ai logo mark.

Color tokens

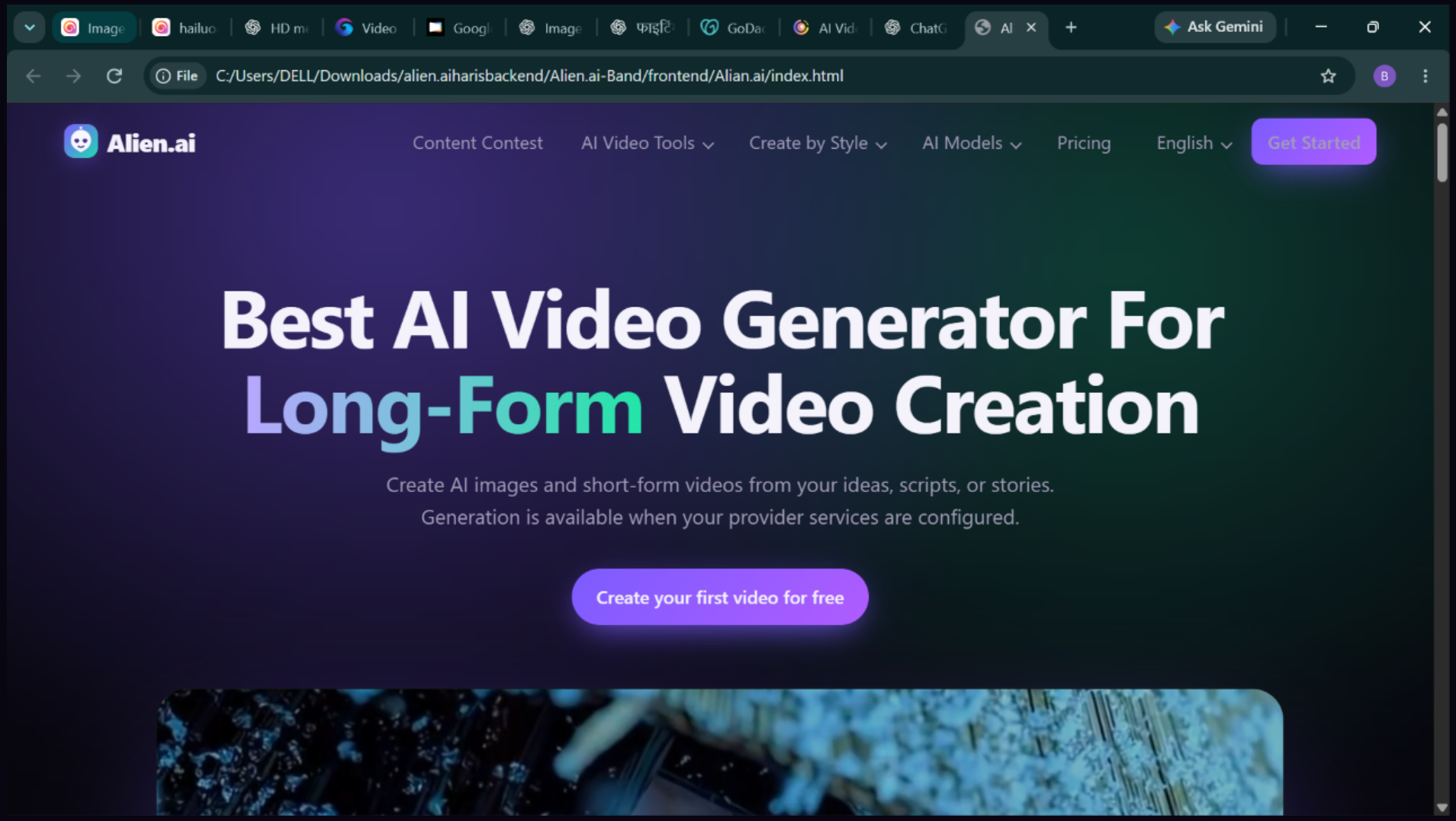
Token	Value	Usage
Background	#07060D / #100D1C	Page background, hero and app shell
Card	#14111F	Panels, menus, history cards
Primary purple	#7C5CFF	Primary buttons, active navigation, focus
Purple glow	#B05CFF	CTA gradient end, glow and hover
Mint green	#22E6A8	Success, progress, subtle highlight
Main text	#F4F2FB	Headings and key content
Muted text	#A29CB8	Descriptions, labels and helper text
Border	rgba(255,255,255,.09)	Subtle separators and card edges

Typography, shapes and motion

- Use a modern sans-serif family such as Inter or system UI. Hero title: bold, oversized, tight line-height; body copy muted and readable.
- Rounded geometry: 14–24 px cards, pill-shaped CTA buttons, subtle 1 px borders and soft shadows.
- Motion: fade/slide-up on section entry, gentle gradient drift in hero background, button hover lift, progress shimmer, skeleton loading and toast transitions.
- Respect prefers-reduced-motion; do not animate essential text in a way that reduces readability.
- Logo direction: compact rounded-square app icon with a friendly alien/planet face; violet-to-mint gradient; white wordmark ‘Alien.ai’.

03 / HOMEPAGE — REFERENCE & BUILD

Reference screenshot supplied by the user. Preserve the visual hierarchy and palette: fixed navigation, centered headline, short supporting copy, primary CTA and a wide featured-media preview below.



Homepage behavior

- Header: Alien.ai logo at left; Content Contest, AI Video Tools, Create by Style, AI Models, Pricing and language controls; Get Started button at right.
- Hero: large two-line headline about AI video creation; short subheading; primary ‘Create your first video for free’ CTA.
- CTA click: if signed out, open /login (signup/sign-in toggle). If already signed in, open /history.
- Below the fold: featured sample video card with play control, title/caption, product feature sections, model/tool highlights, pricing/credits explanation, FAQ and footer.
- Responsive: collapse navigation to a mobile menu; stack sections; maintain readable title and accessible CTA.

04 / SIGNUP, LOGIN & FIRST-LOGIN FLOW

Account identity must be persistent and provider-backed. Do not store raw passwords in a local JSON file, browser storage or plaintext backend file. Use Supabase Auth (email/password) so credentials are securely handled and the same account can sign in on another device.

User journey

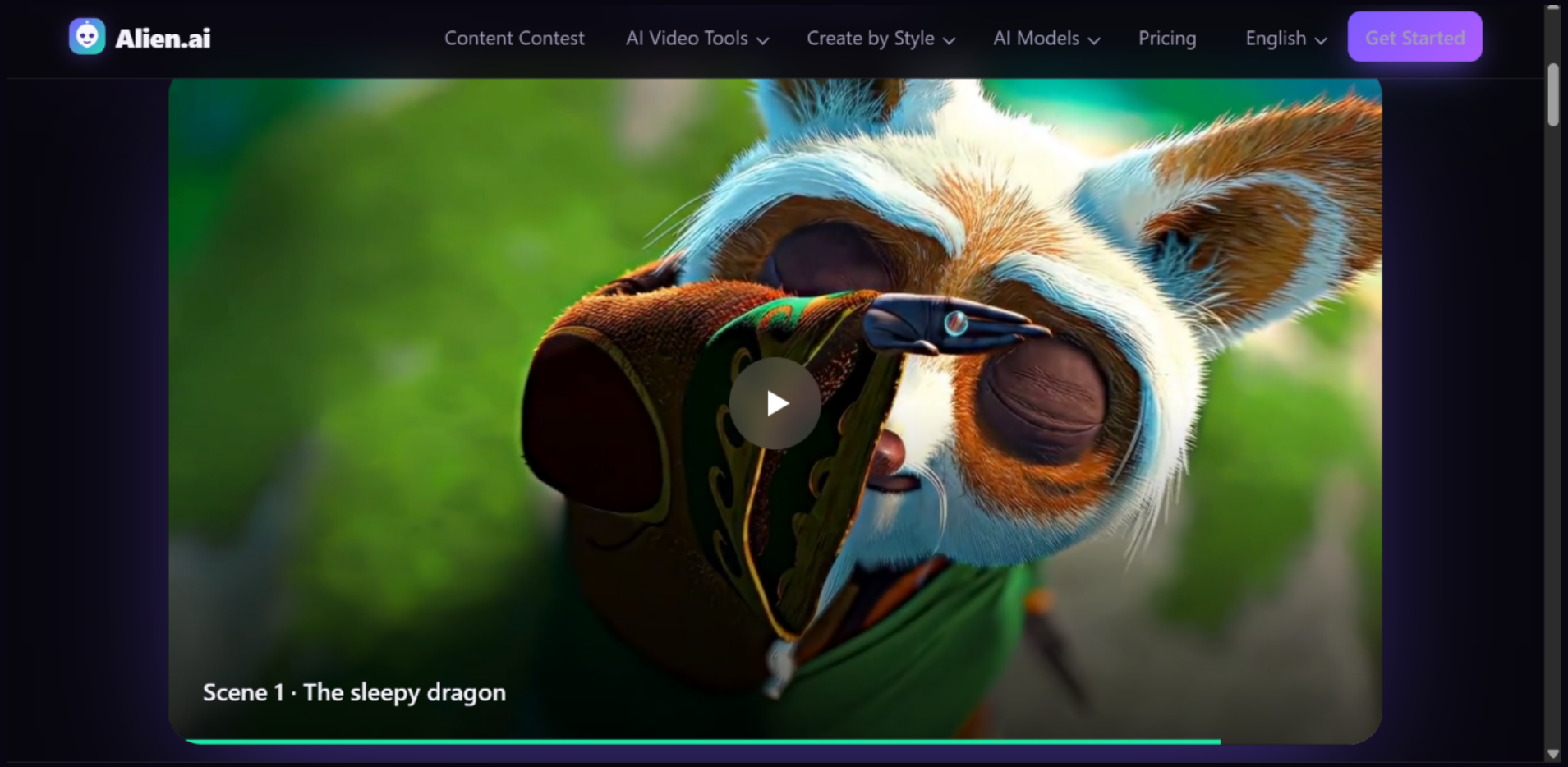
Step	User action	System response
1	Click Get Started / Create	Open Signup/Login screen
2	Choose Create Account	Collect display name, email and password; validate input
3	Submit signup	Supabase Auth creates account; email confirmation policy applies
4	First successful account creation	Create profile and credit account; award exactly 50 credits once
5	Show welcome modal	‘You got 50 credits’ + OK button; do not award credits again on reload
6	Click OK	Open that user’s History page
7	Return on another device	Login with same email/password; fetch same profile, credits and history

Required screens and states

- Signup: display name, email, password, confirm password, terms checkbox and validation messages.
- Login: email, password, show/hide password, forgot password and signup link.
- Loading, invalid credentials, email confirmation pending, network error and session-expired states.
- Use secure, HttpOnly cookies or Supabase session handling; refresh session safely. Add rate limits to auth endpoints.
- Welcome-credit dialog should be triggered by a verified first-time grant event, not merely by opening the page.

05 / HISTORY PAGE & USER LIBRARY

After authentication, the default destination is the account’s History page. This is the user’s private workspace entry point and should make previous work immediately visible.



History page layout and behavior

- Top navigation/header: Alien.ai logo, current credit balance, avatar and account menu.

- Main area: large featured/recent video preview near the top, then a clear centered or prominent ‘Create New’ action.
- Below: History grid/list with thumbnail, prompt snippet, image/video badge, duration, dimensions, created date and status.
- Filters: All, Images, Videos, In Progress, Failed; search by prompt/title; sort newest/oldest.
- Click an item to open detail view with preview, download, copy prompt, regenerate and delete actions.
- All history queries must be filtered by authenticated user ID on the server. User A must never see User B’s media.

06 / CREATION DASHBOARD — GOOGLE FLOW-INSPIRED

Build a creator interface with the same kind of focused workflow shown in Google Flow: a large central workspace, compact left navigation/settings, persistent prompt composer and clear generation status. Keep Alien.ai branding and styling.

Main dashboard regions

Region	Required controls / content
Left sidebar	Alien.ai mark; History; Create; My Media/Uploads; Projects; Settings; help; collapsible rail
Top bar	Project or workspace title; model selector (where supported); credit balance; avatar/profile menu
Main canvas	Generated results, reference uploads, preview player, empty state, progress state and result detail
Prompt composer	Multiline prompt; attach image/video; prompt enhancement optional; Generate button
Composer settings	Image / Video mode; aspect ratio; output quantity for images; video duration: 4s, 5s, 8s, 10s
Result panel	Image grid or video player; download; save to history; regenerate; copy prompt; report issue

Interaction details

- Composer stays anchored near the bottom of the workspace; on small screens it becomes a responsive bottom panel.
- Upload button opens a file picker and supports drag-and-drop. Show thumbnail, filename, remove and upload progress.
- Switching Image/Video updates the settings and estimated credit cost before generation.
- Generate is disabled when prompt is empty, provider is unavailable, upload is incomplete or balance is insufficient.
- Video generation shows queued → processing → completed/failed. Poll task status until terminal state; keep progress visible if user navigates to History.

07 / GENERATION OPTIONS & CREDIT RULES

The requested duration-to-credit schedule must be implemented centrally on the backend and displayed in the UI before the user clicks Generate.

Generation type	Options	Credit charge
Image	OpenAI image size: square, landscape, portrait; quantity selector	5 credits per image output (recommended interpretation; confirm if quantity is charged per output)
Video	4 seconds	5 credits
Video	5 seconds	8 credits
Video	8 seconds	10 credits
Video	10 seconds	15 credits
New account	First verified signup only	+50 credits one time

Credit transaction rules

- Calculate cost server-side from generation type, duration and quantity; never trust a client-submitted credit amount.
- Reserve credits atomically before starting provider generation. Use database transactions/row locks and an idempotency key to prevent double charges.
- If provider generation fails or is cancelled before a billable result, refund the exact reserved amount once and record the refund transaction.
- Show balance before and after generation; show a clear ‘Not enough credits’ message and route to pricing/top-up when needed.
- 50-credit welcome grant must be idempotent and linked to the new authenticated user ID. Never award it again on login or refresh.

Note on image pricing

Your brief specifies variable image charges based on how many images are created but does not provide the exact per-image rate. The ZIP currently documents a default 5-credit image request cost independent of quantity. The implementation should decide and document whether the 5 credits apply per image or per request before launch.

08 / TECH STACK & SYSTEM ARCHITECTURE

The supplied project already uses a TypeScript/Express backend, Supabase, OpenAI image generation and Runway video tasks. Keep this foundation unless there is a deliberate reason to replace it.

Layer	Recommended technology	Responsibility
Frontend	HTML/CSS/JavaScript currently; optional React + TypeScript for scale	Landing page, auth, History, dashboard, animation and responsive UI
Backend API	Node.js 20+ • TypeScript • Express 5	Auth middleware, generation orchestration, uploads, credits, history and validation
Database	Supabase PostgreSQL	Profiles, projects, generations, assets, credit accounts and ledger
Authentication	Supabase Auth (email/password)	Secure signup/login, confirmation, session refresh and password reset
File storage	Supabase Storage (private bucket)	Uploaded references and generated image/video assets
Image provider	OpenAI Images API	Image generation from text prompts
Video provider	Runway API	Text-to-video and image-to-video asynchronous tasks
Validation/security	Zod, Helmet, CORS allowlist, rate limiting	Input validation, HTTP hardening and abuse protection

Request flow

Browser → authenticated Alien.ai API → validate prompt/options → reserve credits in Supabase DB → call OpenAI or Runway → save outputs in Supabase Storage → update generation status and ledger → return signed media URL → refresh History UI.

09 / CODE MAP — WHAT GOES WHERE

File / folder	Purpose / implementation notes
frontend/Alian.ai/index.html	Public homepage; match screenshot design, hero, navigation, feature sections and CTA routing.
frontend/Alian.ai/login.html	Signup/Login UI; connect to real Supabase auth routes; do not rely on demo auth for production.
frontend/Alian.ai/dashboard.html	Creator dashboard; mode selector, prompt composer, uploads, duration/ratio controls and generation results.
frontend/Alian.ai/demo-auth.js	Development/demo helper only. Disable demo mode in production; do not treat demo users as persistent accounts.
src/server.ts / src/app.ts	Start Express server, register middleware and API routes.
src/config.ts	Read environment variables; provider model names and server-side credit settings.
src/routes/auth.ts	Signup, login, refresh, logout, password reset and current-user endpoints.
src/routes/generations.ts	Validate requests, reserve credits, call providers, persist status/results and refund failed tasks.
src/routes/history.ts	Return only the authenticated user’s generation history and signed media URLs.
src/routes/uploads.ts	Validate file MIME/size, upload to private Supabase Storage and return signed URLs.
src/routes/credits.ts	Return balance, cost estimates and transaction history.
src/providers/image.ts	OpenAI Images API adapter.
src/providers/video.ts	Runway task create/status adapter.
src/lib/supabase.ts	Supabase client creation and admin/auth clients.
supabase/migrations/*.sql	Tables, indexes, RLS policies, triggers, credit RPCs and schema updates.

10 / API ENDPOINT PLAN

Method & endpoint	Purpose	Auth
POST /api/auth/supabase/register	Create email/password account	No
POST /api/auth/supabase/login	Sign in and create session	No
POST /api/auth/supabase/refresh	Refresh expired access session	Session
GET /api/auth/me	Read current authenticated user	Yes
POST /api/generations/image	Create image generation request	Yes
POST /api/generations/video	Create Runway video task	Yes
POST /api/generations/image-to-video	Animate uploaded reference image	Yes
GET /api/generations	List user generations/history	Yes
GET /api/generations/:id	Read status and refresh provider task status	Yes
POST /api/uploads	Upload image/video reference	Yes
GET /api/uploads	List user’s uploaded assets	Yes
GET /api/credits	Read current balance and base rates	Yes
GET /api/credits/transactions	Read credit ledger	Yes
GET /api/config/generators	Return enabled models, sizes and durations	No

API contract principles

- Return a consistent JSON envelope: success, data and error. Use correct HTTP status codes.
- Validate duration strictly against 4, 5, 8 and 10 seconds in the UI and backend. Reject unsupported values.
- Use idempotency_key for create requests to protect against duplicate clicks and network retries.
- Store provider job ID and generation status. For async Runway jobs, poll server-side or via authenticated status endpoint until completed/failed.

- Never expose OpenAI, Runway or Supabase service-role secrets in frontend JavaScript.

11 / DATABASE, HISTORY & STORAGE DESIGN

Table / bucket	Key fields	Purpose
profiles	id, display_name, avatar_url	One profile per Supabase auth user
projects	id, user_id, title, description	Optional grouping of generations
generations	id, user_id, provider, model, type, prompt, status, duration, credits_used	One record per generation job
generation_assets	id, generation_id, user_id, storage_path, mime_type, output index	One-to-many outputs for image batches and video files
credit_accounts	user_id, balance	Current balance per account
credit_transactions	user_id, generation_id, type, amount, balance_after, idempotency_key	Auditable credit/debit/refund ledger
Supabase Storage: alien-ai-media	private bucket; path starts with user UUID	Uploaded inputs and generated outputs

Data rules

- Every user-owned table includes user_id and Row Level Security (RLS) policies. Enforce ownership on the backend too.
- Keep storage private; issue short-lived signed URLs only after checking ownership.
- History query: WHERE user_id = authenticated_user_id ORDER BY created_at DESC; paginate rather than loading every item.
- Keep prompt, provider/model, size/aspect ratio, duration, cost, status, timestamps, provider job ID and error details for support/debugging.
- Delete flow should remove database records and associated storage objects according to a clear retention policy.

12 / EXISTING ZIP REVIEW — WHAT IS ALREADY THERE

The supplied ZIP is not an empty starter template. It contains a working project structure and substantial implementation. The points below are based on code and documentation inspected in the archive.

Area	Found in ZIP	Status / action
Backend foundation	TypeScript, Express 5, Node.js, API routes, middleware	Present; run build/tests and verify deployment config
Supabase integration	Auth client, admin client, SQL migrations, RLS, storage bucket	Present; apply migrations to live Supabase and verify health
OpenAI image provider	src/providers/image.ts uses OpenAI Images endpoint	Present; verify key, model access, output storage and quantity pricing
Runway video provider	src/providers/video.ts creates and polls Runway tasks	Present; verify RUNWAY_API_KEY and RUNWAY_VIDEO_MODEL
Auth	Supabase signup/login plus demo-auth paths	Real auth exists; disable demo mode in production
History	generation/history routes and dashboard	Present in backend; test user isolation and asset restoration
Credit system	DB balance, transactions, reserve/refund RPCs	Present; configure 50-credit signup grant and requested duration pricing
Homepage/dashboard	index.html, login.html, dashboard.html	Present; compare every screen to final approved design and complete UI gaps

Critical gaps to verify before calling it launch-ready

- The README says new Supabase users receive 50 credits in the newer migration, while the initial migration shown in the archive creates 100 credits. Apply migrations in the correct order and verify the final trigger grants exactly 50—not 100 or a duplicate.
- The API documentation currently describes 5- and 10-second Runway durations and a default flat 15-credit video charge. Your requested 4/5/8/10-second choices and 5/8/10/15 pricing need code, validation and tests.
- README/API docs currently describe image generation default cost as 5 credits per request, independent of quantity. Decide whether user’s ‘per image’ requirement means 5 credits per output or per request.
- Demo auth is not persistent cross-device storage and demo accounts do not have persistent credits/history. Production must use Supabase Auth.
- Do not claim video/image generation is ready until provider credentials, model access, storage, migrations and health checks have all been verified.

13 / ENVIRONMENT, SECURITY & DEPLOYMENT

Keep secrets in the backend environment only. The ZIP includes a .env file; treat it as sensitive and rotate any real keys that may have been shared or committed. Do not publish it or include it in frontend bundles.

Required environment configuration

```
NODE_ENV=production
PORT=4000
FRONTEND_URL=https://your-domain.example

SUPABASE_URL=...
SUPABASE_PUBLISHABLE_KEY=...
SUPABASE_SECRET_KEY=...
SUPABASE_STORAGE_BUCKET=alien-ai-media

OPENAI_API_KEY=...
OPENAI_IMAGE_MODEL=gpt-image-1

RUNWAY_API_KEY=...
RUNWAY_VIDEO_MODEL=...

IMAGE_CREDIT_COST=5
VIDEO_CREDIT_COST=15
DEMO_AUTH_ENABLED=false
```

Security and release checklist

- Never commit .env or provider keys. Add secret scanning and rotate credentials if exposed.
- Production HTTPS, secure cookies, strict CORS allowlist, Helmet headers, request-size limits and rate limits.
- Validate prompt length, MIME types, upload size, image dimensions and allowed generation parameters.
- Use Supabase RLS and least-privilege policies; service-role key is server-only.
- Protect generation endpoints from abuse and duplicate requests; log request IDs without logging passwords or secret tokens.
- Before launch: npm ci → npm run typecheck → npm run lint → npm test → npm run build → deploy → apply migrations → /api/health check.

14 / END-TO-END TEST PLAN

Test	Expected result
New account signup	One account, one profile, exactly +50 welcome credits
Repeat login / another browser	Same account, same balance and same History records
User A vs User B	No cross-account access to generations, uploads or signed media
Create image	OpenAI result is saved, displayed and restored after refresh
Create 4s / 5s / 8s / 10s video	Correct duration reaches Runway; correct 5/8/10/15 credits reserved
Insufficient balance	Generation blocked before provider call; no negative balance
Provider failure	Generation marked failed; reserved credits refunded exactly once
Double-click / retry	Idempotency prevents duplicate job and duplicate charge
Upload image/video	Only allowed file types/size accepted; file linked to current user
History filters & pagination	Correct records, newest first, stable across refresh
Mobile / tablet / desktop	Responsive nav, composer and media preview remain usable
Reduced motion / keyboard	Accessible controls, focus states and reduced-motion support

Definition of done

A feature is complete only when its UI, API, database behavior, error states, security checks and automated/manual tests all work. A button that only animates or a sample preview that is not connected to a real provider does not count as a finished feature.

15 / STEP-BY-STEP IMPLEMENTATION ROADMAP

Phase	Work	Deliverable / acceptance
1. Audit & stabilize	Review ZIP, remove secrets from source control, confirm migrations and environment config	Clean build; health check; documented provider setup
2. Brand & homepage	Finalize Alien.ai logo, color tokens, hero, CTA, featured video, responsive layout	Homepage matches approved reference
3. Authentication	Connect Supabase signup/login/reset/session; award 50 once	Cross-device login and welcome-credit modal
4. History	Build user-specific history, media previews, filters and Create New	History persists after refresh and login
5. Dashboard UI	Build Flow-inspired workspace, uploads, prompt composer, settings and result panels	Responsive creator dashboard
6. Image pipeline	Connect OpenAI, quantity, sizes, storage, statuses and image credit logic	Real images saved to user library
7. Video pipeline	Runway task creation/status, duration selector, pricing schedule and refunds	Real 4/5/8/10-second videos saved to history
8. QA & launch	Security, accessibility, load/error states, tests, deploy and monitor	Production-ready release checklist passed

Recommended priority

First make signup → 50-credit grant → History → Create New → generation → saved History work end-to-end. Then polish animations, model selectors and marketing sections. This protects the core user journey before investing time in decorative motion.

16 / OPEN DECISIONS BEFORE FINAL BUILD

These items are not fully specified in the supplied brief or conflict with current ZIP defaults. Confirm them before implementing the final billing and generation behavior.

Decision	Current understanding	Recommended action
Image credit pricing	ZIP says 5 credits per request, regardless of quantity; brief says charge based on number of images	Choose 5 credits per generated image OR 5 per batch; update UI/API/ledger consistently
Runway duration support	ZIP docs say 5 and 10 seconds; brief asks 4, 5, 8 and 10 seconds	Verify current Runway model supports all requested durations; do not show unsupported options
Runway billing	Brief sets 4s=5, 5s=8, 8s=10, 10s=15 credits	Implement server-side duration map and automated tests
Welcome credits	Brief says 50; initial SQL snippet shows 100 while later README/migration says 50	Verify final database trigger and existing-account migration; award exactly once
History preview	Brief asks a video preview near top and Create New action	Define empty state and behavior when user has no videos
Image dimensions	User wants image size selector but exact presets not specified	Start with square 1:1, landscape 3:2 and portrait 2:3 if supported by selected OpenAI model

Final product statement

Alien.ai should launch as a reliable AI creation studio: the same visual language across homepage, History and dashboard; one secure account per user; real OpenAI image generation; real Runway video generation; private saved assets; and transparent credit accounting.

END OF BLUEPRINT